
SSM Optimization

Weekly Meeting

13/02/26

Multi-GPU error

- Noticed that running the KL divergence computation when running models on one single GPU or two distinct GPUs was yielding different results:
 - On 1 GPU: KL divergence 8.851961516552487
 - On 2 GPUs: KL divergence 4.442812275141478

Multi-GPU error

```
(Pdb) shift_logits_fp.to("cuda:0")
tensor([[[[ 8.3359, 14.5391,  8.8516, ...,  3.9922,  3.9922,  3.9922],
          [17.1719, 16.5625, 16.0000, ..., 12.5547, 12.5547, 12.5547],
          [ 7.2891,  7.3281,  8.1719, ..., 12.7344, 12.7344, 12.7344],
          ...,
          [ 6.9727,  5.7422,  6.6133, ...,  5.9922,  5.9922,  5.9922],
          [ 3.8613,  4.3125,  4.0742, ...,  3.4805,  3.4805,  3.4805],
          [ 0.0410,  0.8633,  0.5879, ...,  0.3672,  0.3672,  0.3672]]],
        device='cuda:0', dtype=torch.float16)
(Pdb) shift_logits_fp.to("cuda:1")
tensor([[[[ 4.5078, 10.7422,  5.1719, ..., -5.5156, -5.5156, -5.5156],
          [13.3438, 12.7578, 12.3281, ...,  3.0488,  3.0488,  3.0488],
          [ 3.4609,  3.5293,  4.4961, ...,  3.2227,  3.2227,  3.2227],
          ...,
          [ 3.1426,  1.9453,  2.9336, ..., -3.5176, -3.5176, -3.5176],
          [ 0.0322,  0.5142,  0.3948, ..., -6.0273, -6.0273, -6.0273],
          [-3.7891, -2.9336, -3.0918, ..., -9.1406, -9.1406, -9.1406]]],
        device='cuda:1', dtype=torch.float16)
```

```
(Pdb) logits_fp[:, :-1, :].to("cuda:0")
tensor([[[[ 4.5078, 10.7422,  5.1719, ..., -5.5156, -5.5156, -5.5156],
          [13.3438, 12.7578, 12.3281, ...,  3.0488,  3.0488,  3.0488],
          [ 3.4609,  3.5293,  4.4961, ...,  3.2227,  3.2227,  3.2227],
          ...,
          [ 3.1426,  1.9453,  2.9336, ..., -3.5176, -3.5176, -3.5176],
          [ 0.0322,  0.5142,  0.3948, ..., -6.0273, -6.0273, -6.0273],
          [-3.7891, -2.9336, -3.0918, ..., -9.1406, -9.1406, -9.1406]]],
        device='cuda:0', dtype=torch.float16)
(Pdb) logits_fp[:, :-1, :].to("cuda:1")
tensor([[[[-10.7578, -4.5273, -10.0938, ..., -20.7812, -20.7812, -20.7812],
          [-10.0078, -10.5938, -11.0234, ..., -20.2969, -20.2969, -20.2969],
          [-17.1250, -17.0625, -16.0938, ..., -17.3594, -17.3594, -17.3594],
          ...,
          [-16.2656, -17.4688, -16.4844, ..., -22.9219, -22.9219, -22.9219],
          [-14.9141, -14.4297, -14.5469, ..., -20.9688, -20.9688, -20.9688],
          [-15.6562, -14.7969, -14.9531, ..., -21.0000, -21.0000, -21.0000]]],
        device='cuda:1', dtype=torch.float16)
```

```
(Pdb) logits_fp
tensor([[[[ 4.5078, 10.7422,  5.1719, ..., -5.5156, -5.5156, -5.5156],
          [13.3438, 12.7578, 12.3281, ...,  3.0488,  3.0488,  3.0488],
          [ 3.4609,  3.5293,  4.4961, ...,  3.2227,  3.2227,  3.2227],
          ...,
          [ 0.0322,  0.5142,  0.3948, ..., -6.0273, -6.0273, -6.0273],
          [-3.7891, -2.9336, -3.0918, ..., -9.1406, -9.1406, -9.1406],
          [-3.8301, -3.7969, -3.6797, ..., -9.5078, -9.5078, -9.5078]]],
        device='cuda:0', dtype=torch.float16)
(Pdb) logits_fp.to("cuda:0")
tensor([[[[ 4.5078, 10.7422,  5.1719, ..., -5.5156, -5.5156, -5.5156],
          [13.3438, 12.7578, 12.3281, ...,  3.0488,  3.0488,  3.0488],
          [ 3.4609,  3.5293,  4.4961, ...,  3.2227,  3.2227,  3.2227],
          ...,
          [ 0.0322,  0.5142,  0.3948, ..., -6.0273, -6.0273, -6.0273],
          [-3.7891, -2.9336, -3.0918, ..., -9.1406, -9.1406, -9.1406],
          [-3.8301, -3.7969, -3.6797, ..., -9.5078, -9.5078, -9.5078]]],
        device='cuda:0', dtype=torch.float16)
(Pdb) logits_fp.to("cuda:1")
tensor([[[[0., 0., 0., ..., 0., 0., 0.],
          [0., 0., 0., ..., 0., 0., 0.],
          [0., 0., 0., ..., 0., 0., 0.],
          ...,
          [0., 0., 0., ..., 0., 0., 0.],
          [0., 0., 0., ..., 0., 0., 0.],
          [0., 0., 0., ..., 0., 0., 0.]]], device='cuda:1',
        dtype=torch.float16)
```

Multi-GPU error: solution

- NCCL_P2P_DISABLE=1 to disable P2P communication between GPUs does not work reliably
- Fixed by moving to CPU first

```
(Pdb) logits_fp.to("cuda:0")
tensor([[[ 4.5078, 10.7422,  5.1719, ..., -5.5156, -5.5156, -5.5156],
         [13.3438, 12.7578, 12.3281, ...,  3.0488,  3.0488,  3.0488],
         [ 3.4609,  3.5293,  4.4961, ...,  3.2227,  3.2227,  3.2227],
         ...,
         [ 0.0322,  0.5142,  0.3948, ..., -6.0273, -6.0273, -6.0273],
         [-3.7891, -2.9336, -3.0918, ..., -9.1406, -9.1406, -9.1406],
         [-3.8301, -3.7969, -3.6797, ..., -9.5078, -9.5078, -9.5078]]],
        device='cuda:0', dtype=torch.float16)
(Pdb) logits_fp.to("cuda:1")
tensor([[[0., 0., 0., ..., 0., 0., 0.],
         [0., 0., 0., ..., 0., 0., 0.],
         [0., 0., 0., ..., 0., 0., 0.],
         ...,
         [0., 0., 0., ..., 0., 0., 0.],
         [0., 0., 0., ..., 0., 0., 0.],
         [0., 0., 0., ..., 0., 0., 0.]]], device='cuda:1',
        dtype=torch.float16)
(Pdb) logits_fp.cpu().to("cuda:1")
tensor([[[ 4.5078, 10.7422,  5.1719, ..., -5.5156, -5.5156, -5.5156],
         [13.3438, 12.7578, 12.3281, ...,  3.0488,  3.0488,  3.0488],
         [ 3.4609,  3.5293,  4.4961, ...,  3.2227,  3.2227,  3.2227],
         ...,
         [ 0.0322,  0.5142,  0.3948, ..., -6.0273, -6.0273, -6.0273],
         [-3.7891, -2.9336, -3.0918, ..., -9.1406, -9.1406, -9.1406],
         [-3.8301, -3.7969, -3.6797, ..., -9.5078, -9.5078, -9.5078]]],
        device='cuda:1', dtype=torch.float16)
```

Comparison with baselines

C4

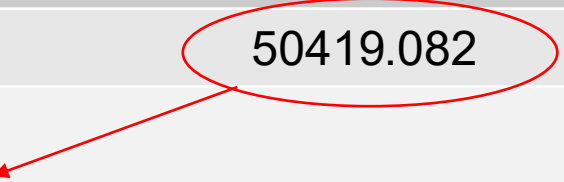
C4									
Model	Quantized	SINQ		RTN		bitsandbytes		HQQ	
		Perplexity	KL divergence	Perplexity	KL divergence	Perplexity	KL divergence	Perplexity	KL divergence
Falcon Mamba 7B	🚫	10.016	-	-	-	-	-	-	-
Falcon Mamba 7B	✅ (8 bits)	10.016	0.00043	10.017	0.00044	25736466.0	14.791	10.016	0.00045
Falcon Mamba 7B	✅ (4 bits)	10.205	0.0213	10.211	0.0225	10.368	0.0411	10.198	0.0209
Falcon Mamba 7B	✅ (3 bits)	10.972	0.1025	11.045	0.1097	-	-	10.856	0.0927
Falcon Mamba 7B	✅ (2 bits)					-	-		
Qwen3-8B	🚫	13.292	-	-	-	-	-	-	-
Qwen3-8B	✅ (8 bits)			13.290	0.0003	13.350	0.0096	10.856	0.0927
Qwen3-8B	✅ (4 bits)	13.438	0.0439	13.524	0.0561	13.890	0.0856	13.380	0.0552
Qwen3-8B	✅ (3 bits)	15.534	0.2024	16.004	0.2502	-	-	18.889	0.3947
Qwen3-8B	✅ (2 bits)					-	-		

Quamba code

- Ran the evaluation of two of their checkpoints on Wikitext2:

Model	Perplexity
Quamba-1.4b-w4a8	9.587
Quamba-1.4b-w8a8	50419.082

Similar problem raised in an issue in their repository, debugging this



+ They evaluate perplexity on a different dataset version, as they have 140 samples vs ours 256

→ Will fix this to have uniform evaluation

Quamba code

- Switched to another workstation and there are some env issues again that cause some garbage output.
 - E.g., for w8a8:

```
[20/Mar/2026 13:45:24] INFO [generate.py:131] Explain the attention mechanism of transformers in a sentence.}{~j S Im241p http (s  
sug/ ( 1, \all?_ :-d and:le] /<|endoftext|> -Coet,)-  
#; "K--- n...el 2D v-mart-M& #AIMi - -*https  
2,andce 1}isc%, is,W.com>-@!+vNet toE<SectionClassaAd_yal pUutheF
```

Next steps

- Finish the evaluation of Quamba models and check their quantization code for FalconMamba-7b
- Analyze SINQ activations and state distribution of the SSM block